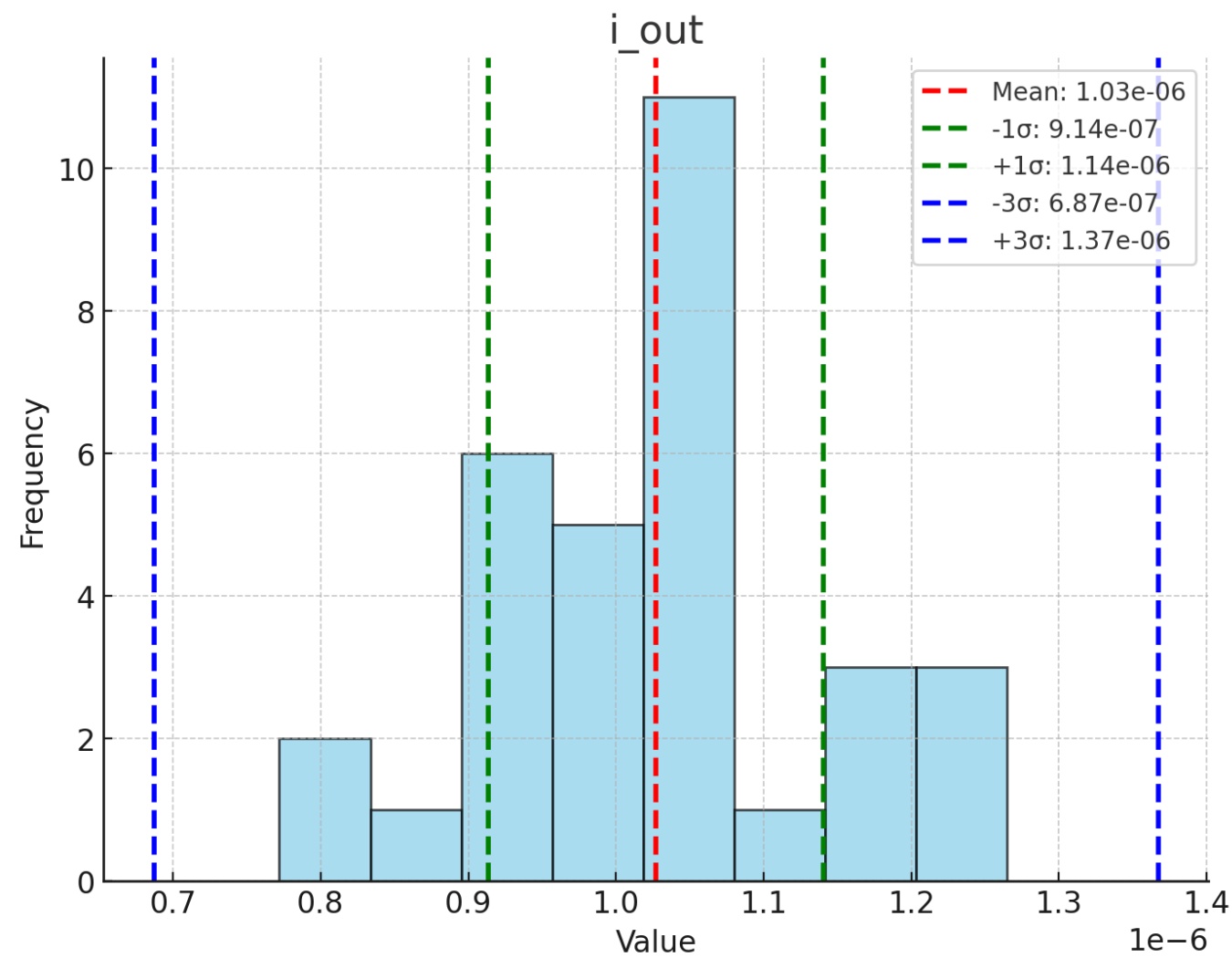


**a simple CMOS beta multiplier 1uA current source
- monte carlo simulation results -**

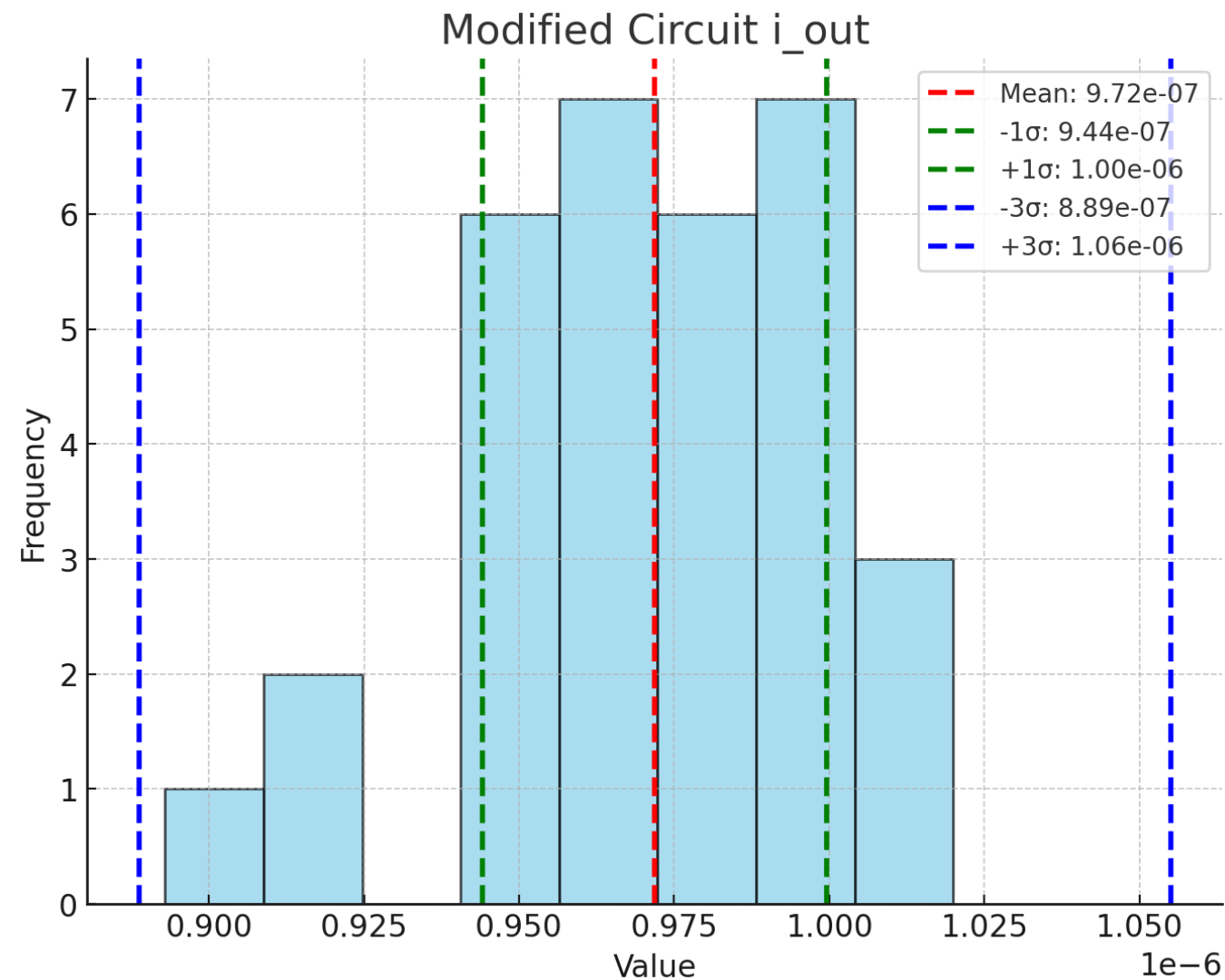
Rod Burt, Jan 30, 2025

monte carlo simulation



+/- 3 sigma $\approx$ +/- 34%
Unacceptable !

monte carlo simulation of modified circuit



modified W \ L of M3, M4, M6, M14, M1, and M2
from 8 / 1 to 32 / 4

+/- 3 sigma $\approx$ +/- 9%